

WEAK OKA-1 DOMAINS WITHOUT THE OKA-1 PROPERTY

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ABSTRACT. For every $n \geq 2$, we construct a noncompact domain $X_n \subset \mathbb{C}^n$ that is weak Oka-1 but not Oka-1. For every compact set $K \subset \mathbb{C}$ with connected complement, every holomorphic map from a neighbourhood of K into X_n can be approximated uniformly on K by polynomial maps $\mathbb{C} \rightarrow \mathbb{C}^n$ whose images lie in X_n . Nevertheless, every entire curve in X_n through a certain point is constant, so even approximation with one prescribed value can fail. The obstruction is provided by a bounded plurisubharmonic function. The corresponding question for compact complex manifolds remains open.

1. INTRODUCTION

Alarcón and Forstnerič [1] introduced the Oka-1 property, which requires approximation and discrete jet interpolation for maps from open Riemann surfaces. Xie [11, Definition 1.7] introduced the weak Oka-1 property, formulated in terms of approximation on two disjoint discs. Chen–Fornæss–Xie [2, Proposition 3.1] proved that it is equivalent to uniform approximation of maps holomorphic near a closed disc by entire curves. Every Oka-1 manifold is weak Oka-1. For background on Oka theory, see Forstnerič [4]; further examples and structural properties of Oka-1 manifolds are developed by Forstnerič–Lárusson [6].

Xie [11, Question 3.4] asked whether there is a compact weak Oka-1 manifold that is not Oka-1. Equivalently, one may ask the following question, which remains open.

Question 1.1. For a connected compact complex manifold, is the weak Oka-1 property equivalent to the Oka-1 property?

The unresolved direction is $\text{weak Oka-1} \implies \text{Oka-1}$. A positive answer would give, for compact targets and one-dimensional sources, an analogue of Forstnerič’s theorem [5, Theorem 0.1] that Runge approximation on convex compact sets in all affine source dimensions implies the Oka property. Chen–Fornæss–Xie [2, Question 5.3] subsequently asked whether the implication holds for arbitrary complex manifolds, without a compactness assumption. We answer this unrestricted question negatively; the compact case above remains open.

Theorem 1.2. *For every integer $n \geq 2$, there exist an unbounded domain $X_n \subset \mathbb{C}^n$ and a point $a \in X_n$ with the following properties.*

- (i) *For every compact set $K \subset \mathbb{C}$ with connected complement, every holomorphic map from a neighbourhood of K to X_n can be approximated uniformly on K by polynomial maps $\mathbb{C} \rightarrow \mathbb{C}^n$ whose images lie in X_n . In particular, X_n is weak Oka-1.*
- (ii) *Every entire curve $\mathbb{C} \rightarrow X_n$ passing through a is constant. In particular, X_n is not Oka-1.*

The construction starts with the union $A \subset \mathbb{C}^n$ of the images of polynomial maps with coefficients in $\mathbb{Q} + i\mathbb{Q}$. This set is dense and path-connected. Since $n \geq 2$, every such image lies in the zero set of a nonzero nonconstant polynomial over $\mathbb{Q} + i\mathbb{Q}$. The countable union Z of all these zero sets is pluripolar and contains A . Josefson’s theorem [7] (see also [8, Theorem 3.3.20]) yields $u \in \text{PSH}(\mathbb{C}^n)$

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and $a \in \mathbb{C}^n$ such that $Z \subset \{u = -\infty\}$ and $u(a) = 0$. We then take $X_n = \{u < 1\}$. Runge approximation followed by a small perturbation of the coefficients gives polynomial approximants whose entire images are contained in A . The bounded plurisubharmonic function $v = \max\{u, -1\}|_{X_n}$ is constant along every entire curve, but has different values on A and at a . It therefore prevents two-point interpolation, the Liouville obstruction noted by Alarcón and Forstnerič [1, Introduction]. Since the pole set contains the hyperplanes $\{z_\ell = 0\}$ and $\{z_\ell = 1\}$, the little Picard theorem gives the stronger conclusion in part (ii).

The example does not resolve [Question 1.1](#): on a connected compact complex manifold every plurisubharmonic function is constant, so the separator used here has no direct counterpart in the compact setting.

2. DEFINITIONS AND CONSTRUCTION OF THE DOMAIN

A domain is a connected open set. A distance is *compatible* if it induces the given topology. Every connected complex manifold admits a complete compatible distance [9, Theorem 1].

Definition 2.1. A connected complex manifold X with a complete compatible distance d_X is *weak Oka-1* if the following holds for every pair of disjoint closed Euclidean discs D_1, D_2 contained in a closed Euclidean disc $D \subset \mathbb{C}$. For every neighbourhood U of $D_1 \cup D_2$, every $f \in \mathcal{O}(U, X)$ and every $\epsilon > 0$, there are a neighbourhood $\hat{U}$ of D and $F \in \mathcal{O}(\hat{U}, X)$ such that

$$\sup_{z \in D_1 \cup D_2} d_X(F(z), f(z)) < \epsilon.$$

The property in [Definition 2.1](#) is independent of the compatible distance. For later use, we record the more general uniform comparison on compact subsets. For two compatible distances d, d' and a compact set $C \subset X$, every $\epsilon > 0$ admits $\delta > 0$ such that

$$x \in C, y \in X, d(x, y) < \delta \implies d'(x, y) < \epsilon. \quad (1)$$

Otherwise there would be $x_j \in C$ and $y_j \in X$ with $d(x_j, y_j) \rightarrow 0$ and $d'(x_j, y_j) \geq \epsilon$. A subsequence satisfies $x_j \rightarrow x \in C$, hence $y_j \rightarrow x$, contradicting compatibility. Applying (1) to $C = f(D_1 \cup D_2)$ and interchanging d, d' proves the assertion.

Definition 2.2. A connected complex manifold X with a compatible distance d_X is *Oka-1* if the following holds. For an open Riemann surface S , a compact $\mathcal{O}(S)$ -convex set $K \subset S$, a closed discrete set $E \subset S$ and a function $m : E \rightarrow \{1, 2, \dots\}$, suppose that $f : S \rightarrow X$ is continuous and holomorphic near $K \cup E$. For every $\epsilon > 0$, there is a holomorphic map $F : S \rightarrow X$, homotopic to f , such that

- (i) $\sup_{z \in K} d_X(F(z), f(z)) < \epsilon$;
- (ii) F and f have the same $(m(p) - 1)$ -jet at every $p \in E$.

Here compact $\mathcal{O}(S)$ -convex sets are precisely compact Runge sets: $S \setminus K$ has no relatively compact connected component. In [Definition 2.2\(ii\)](#), derivatives of orders strictly less than $m(p)$ agree in local coordinates; $m(p) = 1$ prescribes the value. For $m(p) \geq 2$, the prescribed jet order is $m(p) - 1$ in the convention of [1, Definition 1.1]. Their first-order interpolation also gives the value-only case $m(p) = 1$, so the two formulations are equivalent.

In particular, any two points x_0, x_1 in a connected Oka-1 manifold lie on an entire curve F with $F(0) = x_0$ and $F(1) = x_1$. This follows directly from [Definition 2.2](#): path-connectedness gives a continuous map $\mathbb{C} \rightarrow X$ equal to x_0 near 0 and to x_1 near 1, to which one applies the interpolation property with $S = \mathbb{C}$, $K = \emptyset$, $E = \{0, 1\}$ and $m(0) = m(1) = 1$.

We allow plurisubharmonic functions to take the value $-\infty$ but not $+\infty$, and exclude the identically $-\infty$ function from PSH. For a nonzero holomorphic function g , $\log |g|$ is plurisubharmonic [3, Chapter I, Example 5.12]. Finite maxima of plurisubharmonic functions are plurisubharmonic [3, Chapter I, Theorem 5.6]. For an arbitrary holomorphic map $F : Y \rightarrow \Omega$ from a connected Riemann

surface and $w \in \text{PSH}(\Omega)$, the composition $w \circ F$ is subharmonic or identically $-\infty$ by the holomorphic pullback theorem [3, Chapter I, Theorem 5.11]. Finally, a subharmonic function on $\mathbb{C}$ bounded above is constant [10, Corollary 2.3.4]. A subset of $\mathbb{C}^n$ is *pluripolar* if, near each of its points, it is contained in the $-\infty$ locus of a plurisubharmonic function on a connected neighbourhood that is not identically $-\infty$.

We now construct the domain. Fix $n \geq 2$. The field $\mathbf{k} = \mathbb{Q} + i\mathbb{Q}$ is countable and dense in $\mathbb{C}$, so the family $\mathcal{P} = \mathbf{k}[t]^n$ of polynomial maps is countable. Its images have union

$$A = \bigcup_{\gamma \in \mathcal{P}} \gamma(\mathbb{C}) \subset \mathbb{C}^n. \quad (2)$$

Lemma 2.3. *The union A and the polynomial maps in $\mathcal{P}$ have the following properties.*

- (i) *The set A is dense and path-connected, and contains $\mathbb{C} \times \{0\}^{n-1}$.*
- (ii) *Every $\gamma \in \mathcal{P}$ satisfies $P \circ \gamma = 0$ for some nonzero nonconstant $P \in \mathbf{k}[z_1, \dots, z_n]$.*

Proof. (i) Constant maps give $\mathbf{k}^n \subset A$, so A is dense. For $x = \gamma(t_0) \in A$, the path $s \mapsto \gamma((1-s)t_0)$ joins x to $\gamma(0) \in \mathbf{k}^n$. For each $b \in \mathbf{k}^n$, the map $t \mapsto tb$ belongs to $\mathcal{P}$ and joins 0 to b on $[0, 1]$. Thus A is path-connected and contains the stated line.

(ii) The coordinates of γ lie in $\mathbf{k}(t)$, which has transcendence degree one over $\mathbf{k}$. Since $n \geq 2$, these coordinates are algebraically dependent. A nonzero polynomial relation cannot be constant, proving the last assertion. $\square$

Enumerate the nonzero nonconstant polynomials in $\mathbf{k}[z_1, \dots, z_n]$ as $P_1, P_2, \dots$. Each zero set $P_j^{-1}(0)$ is pluripolar, since it is the $-\infty$ locus of $\log |P_j|$. The countable union

$$Z = \bigcup_{j \geq 1} P_j^{-1}(0) \quad (3)$$

is therefore pluripolar by [8, Proposition 3.3.21], and $A \subset Z$ by Lemma 2.3. Josefson's theorem [7] (see also [8, Theorem 3.3.20]) gives a function $w \in \text{PSH}(\mathbb{C}^n)$, $w \not\equiv -\infty$, such that $Z \subset \{w = -\infty\}$. Choose $a \in \mathbb{C}^n$ with $w(a) > -\infty$. Normalization gives

$$u = w - w(a) \in \text{PSH}(\mathbb{C}^n), \quad u(a) = 0, \quad Z \subset \{u = -\infty\}. \quad (4)$$

In particular, $a \notin Z$ and $u|_A = -\infty$.

The strict sublevel set

$$X_n = \{z \in \mathbb{C}^n : u(z) < 1\} \quad (5)$$

contains A and a .

Proposition 2.4. *The set X_n is an unbounded domain in $\mathbb{C}^n$.*

Proof. Upper semicontinuity of u gives openness. Each connected component of X_n is open and hence meets the dense set A . Since A is connected, it lies in a single component, so X_n is connected. The complex line in Lemma 2.3 lies in X_n , which is therefore unbounded and noncompact. $\square$

3. APPROXIMATION AND INTERPOLATION

Theorem 3.1. *For a compact set $K \subset \mathbb{C}$ with connected complement and an open neighbourhood U of K , suppose that $f \in \mathcal{O}(U, X_n)$. For every compatible distance d_{X_n} and every $\epsilon > 0$, there is $\gamma \in \mathbf{k}[t]^n$ such that*

$$\gamma(\mathbb{C}) \subset A \subset X_n, \quad \sup_{z \in K} d_{X_n}(\gamma(z), f(z)) < \epsilon.$$

Proof. For $K = \emptyset$ any constant map into A suffices. Assume that $K \neq \emptyset$. Apply (1) to the Euclidean distance on X_n , the distance d_{X_n} and the compact set $f(K)$ to obtain $\delta > 0$. The polynomial form of Runge's theorem [2, Theorem 2.1], applied coordinatewise with error $\delta/(2\sqrt{n})$, gives a polynomial map $q : \mathbb{C} \rightarrow \mathbb{C}^n$ with $\sup_K \|q - f\| < \delta/2$.

Write $q_\ell(t) = \sum_{r=0}^N b_{\ell r} t^r$ for $1 \leq \ell \leq n$ and $R = \max\{1, \sup_K |z|\}$. Density of $\mathbf{k}$ permits coefficients $\tilde{b}_{\ell r} \in \mathbf{k}$ satisfying

$$\sum_{\ell=1}^n \sum_{r=0}^N |\tilde{b}_{\ell r} - b_{\ell r}| R^r < \frac{\delta}{2}.$$

The polynomial map γ with coordinates $\gamma_\ell(t) = \sum_{r=0}^N \tilde{b}_{\ell r} t^r$ then has $\sup_K \|\gamma - q\| < \delta/2$, so $\sup_K \|\gamma - f\| < \delta$. Its full image lies in A by (2), and (1) gives the required estimate. $\square$

Since the complement of two disjoint closed discs is connected, [Theorem 3.1](#) applies to their union. The resulting entire map satisfies [Definition 2.1](#) with $\hat{U} = \mathbb{C}$. Thus X_n is weak Oka-1.

The function

$$v = \max\{u, -1\}|_{X_n} \quad (6)$$

is finite and plurisubharmonic, with $-1 \leq v < 1$, $v|_A = -1$ and $v(a) = 0$. It is not continuous at a : any sequence $a_j \in A$ converging to a has $v(a_j) = -1$. Uniform approximation by curves in A therefore does not preserve the value of v at a limit point.

Proposition 3.2. *The function v is constant along every entire curve in X_n . There is no entire curve $F : \mathbb{C} \rightarrow X_n$ with $F(0) = 0$ and $F(1) = a$; in particular, X_n is not Oka-1.*

Proof. For an entire curve $F : \mathbb{C} \rightarrow X_n$, the holomorphic pullback theorem makes $v \circ F$ subharmonic. It is finite and bounded above by 1, so Liouville's theorem makes it constant. Since $0 \in A$, the specified values would give $(v \circ F)(0) = -1$ and $(v \circ F)(1) = 0$, a contradiction. The two-point consequence of the Oka-1 property recorded above then proves that X_n is not Oka-1. The failure of Oka-1 also follows from the Liouville property of Oka-1 manifolds [[1](#), Corollary 7.5], since $v - 1$ is a nonconstant negative plurisubharmonic function on X_n . $\square$

Proposition 3.3. *Every nonconstant entire curve in X_n has image in $\{u = -\infty\}$. Consequently, every entire curve passing through a is constant.*

Proof. For $F : \mathbb{C} \rightarrow X_n$, the function $u \circ F$ is either identically $-\infty$ or subharmonic and bounded above. Assume that $u \circ F \not\equiv -\infty$. Then $u \circ F$ is a finite constant by Liouville's theorem. For each ℓ , the polynomials z_ℓ and $z_\ell - 1$ occur in the enumeration, and (4) places their zero sets in $\{u = -\infty\}$. Hence the coordinate function F_ℓ omits both 0 and 1. The little Picard theorem makes every F_ℓ constant. Thus every nonconstant entire curve satisfies $u \circ F \equiv -\infty$, proving the first assertion. The second follows from $u(a) = 0$. $\square$

Proof of Theorem 1.2. (i) The domain is unbounded by [Proposition 2.4](#); its polynomial approximation property and weak Oka-1 property follow from [Theorem 3.1](#) and its two-disc application. (ii) The assertion about curves through a is [Proposition 3.3](#), and the failure of Oka-1 is [Proposition 3.2](#). $\square$

Corollary 3.4. *There is a map f , holomorphic near $\overline{\mathbb{D}}$ with values in X_n and $f(0) = a$, which cannot be approximated uniformly on $\overline{\mathbb{D}}$ by entire curves $F : \mathbb{C} \rightarrow X_n$ satisfying $F(0) = a$.*

Proof. Since X_n is open, some $\rho > 0$ satisfies $a + \rho t e_1 \in X_n$ for $|t| < 2$, where $e_1 = (1, 0, \dots, 0)$. For $f(t) = a + \rho t e_1$, [Proposition 3.3](#) forces every such F to be the constant a . Thus $\sup_{\overline{\mathbb{D}}} \|F - f\| = \rho > 0$. By the compact-set distance comparison, the failure of approximation is independent of the chosen compatible distance. $\square$

The discontinuity of v is essential to this separation, as the following observation shows.

Proposition 3.5. *Every bounded-above continuous plurisubharmonic function on a connected weak Oka-1 manifold is constant.*

Proof. Let X be a connected weak Oka-1 manifold and $w \in \text{PSH}(X)$ a continuous function bounded above. For a neighbourhood U of $\overline{\mathbb{D}}$ and a map $g \in \mathcal{O}(U, X)$, the entire-curve characterization [2, Proposition 3.1] gives entire curves $F_j : \mathbb{C} \rightarrow X$ converging uniformly to g on $\overline{\mathbb{D}}$. Each $w \circ F_j$ is subharmonic or identically $-\infty$ and bounded above, hence constant by Liouville's theorem. For every $z \in \mathbb{D}$, continuity gives

$$w(g(z)) = \lim_{j \rightarrow \infty} w(F_j(z)) = \lim_{j \rightarrow \infty} w(F_j(0)) = w(g(0)).$$

For any coordinate ball and any two points in a smaller concentric ball, a small affine disc in these coordinates passes through both points. The equality above therefore shows that w is locally constant, hence constant on X . $\square$

The dimension restriction also has a direct explanation: for $n = 1$ the map $t \mapsto t$ belongs to $\mathcal{P}$, so $A = \mathbb{C}$ and there can be no point a with $u(a) = 0$ while $u|_A = -\infty$. For compact targets, Question 1.1 remains open. The domains constructed here show that approximation alone does not imply Oka-1 without compactness; they neither prove nor disprove the equivalence of weak Oka-1 and Oka-1 for compact complex manifolds.

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